

New Wheels Project

Introduction to SQL

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List of contents	Page no
1. Problem Statement	4
1.1 Context	4
1.2 Objective	4
1.3 Data Description	4
1.4 Data Dictionary	5
1.5 ER Diagram	6
2. Business Questions	6
2.1 Question 1	6
2.2 Question 2	10
2.3 Question 3	12
2.4 Question 4	15
2.5 Question 5	19
2.6 Question 6	23
2.7 Question 7	25
2.8 Question 8	27
2.9 Question 9	29
2.10 Question 10	31
3. Business Metrics Overview	33
4. Business Recommendations	34

List of figures	Page No
Fig 1.5 - ER diagram for New Wheels	6
Fig 2.1.1 - Total Number of Customers placed orders	7
Fig 2.1.2 - Customer Distribution by State	9
Fig 2.2 - Top Vehicle Makers by Customer Count	11
Fig 2.3.1 - Most Popular Vehicle Maker by State(Query)	13
Fig 2.3.2 - Most Popular Vehicle Maker by State	14
Fig 2.4.1 - Overall Average Customer Rating	16
Fig 2.4.2 - Average rating in each quarter	18
Fig 2.5.1 - Percentage distribution of feedback(Query)	21
Fig 2.5.2 - Percentage distribution of feedback	22
Fig 2.6 - Quarterly Orders	24
Fig 2.7.1 - Quarterly Revenue(Query)	26
Fig 2.7.2 - Quarterly Revenue	26
Fig 2.8 - Quarterly Orders and Total Net Revenue	28
Fig 2.9 - Average Discount by Credit Card Type	30
Fig 2.10 - Average Shipping Days by Quarter	32
Fig 3 - Key metrics	33

1. Problem Statement

1.1 Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

1.2 Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

1.3 Data Description

The data provided has

Attributes on the vehicles New-Wheels sells - What are the make, model, and year? What is the price point?

Attributes on the customers, such as where they live and payment methods

Attributes on orders and shipments, such as when the order was shipped and received, what the after-sales feedback was, and so on.

1.4 Data Dictionary

shipper_id: Unique ID of the Shipper
shipper_name: Name of the Shipper
shipper_contact_details: Contact detail of the Shipper
product_id: Unique ID of the Product
vehicle_maker: Vehicle Manufacturing company name
vehicle_model: Vehicle model name
vehicle_color: Color of the Vehicle
vehicle_model_year: Year of Manufacturing
vehicle_price: Price of the Vehicle
quantity: Ordered Quantity
customer_id: Unique ID of the customer
customer_name: Name of the customer
gender: Gender of the customer
job_title: Job Title of the customer
phone_number: Contact detail of the customer
email_address: Email address of the customer
city: Residing city of the customer
country: Residing country of the customer
state: Residing state of the customer
customer_address: Address of the customer
order_date: Date on which customer ordered the vehicle
order_id: Unique ID of the order
ship_date: Shipment Date
ship_mode: Shipping Mode/Class
shipping: Shipping Ways
postal_code: Postal Code of the customer
discount: Discount given to the customer for the particular order by credit card in percentage
credit_card_type: Credit Card Type
credit_card_number: Credit card number
customer_feedback: Feedback of the customer
quarter_number : Quarter Number

1.5 ER Diagram

Entity-Relationship Diagram

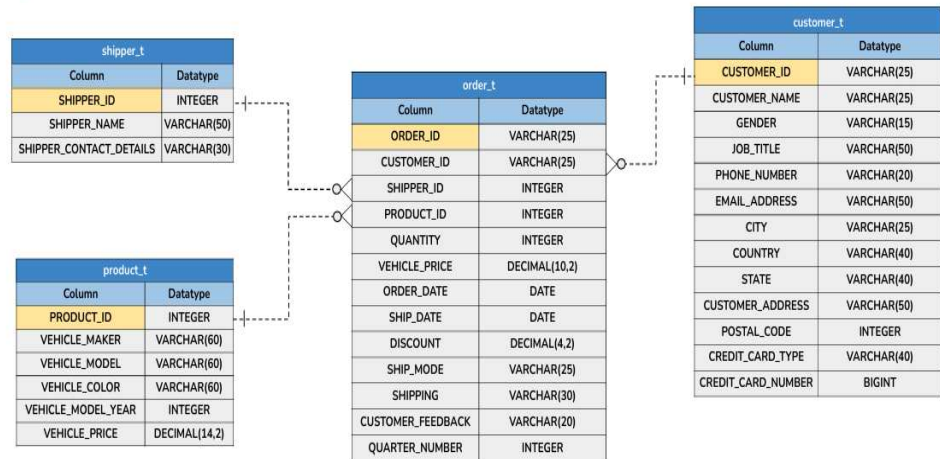


Fig 1.5 - ER diagram for New Wheels

2. Business Questions

2.1 Question 1

Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Total customers who placed orders

Solution Query:

```
SELECT
    COUNT(DISTINCT customer_id) AS
total_customers
FROM order_t;
```

Output:

Result: **Passed**

✓ Query 1

Query:

```
SELECT  
    COUNT(DISTINCT customer_id) AS total_customers  
FROM order_t
```

Output:

Showing 1 rows

total_customers
994

Fig 2.1.1 - Total Number of Customers placed orders

Observations and insights:

- There are 994 different customers who have placed at least one order.
- The business has served 994 unique customers so far.

Distribution of customers across states

Solution Query:

```
SELECT
    c.state,
    COUNT(DISTINCT c.customer_id) AS
customer_count
FROM customer_t c
JOIN order_t o USING (customer_id)
GROUP BY c.state
ORDER BY customer_count DESC;
```

Output:

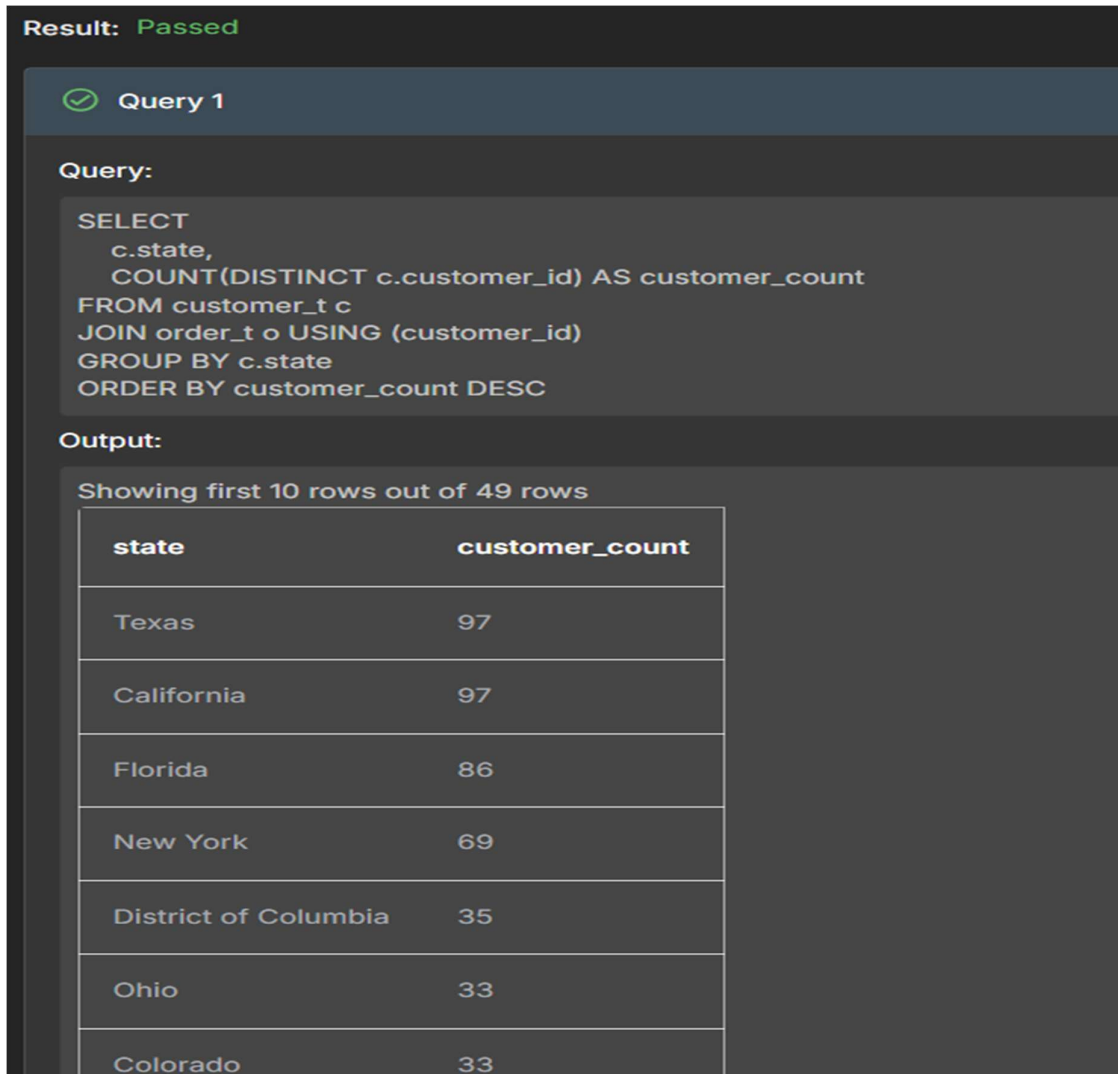


Fig 2.1.2 - Customer Distribution by State

Observations and insights:

- The number of customers who placed orders varies across states, with some states having significantly higher customer counts than others.
- Top states in the output:
 - **Texas – 97 customers**
 - **California – 97 customers**
 - **Florida – 86 customers**
 - **New York – 69 customers**
 - **District of Columbia – 35 customers**
- Texas and California have the highest number of purchasing customers (97 each). These states represent major demand centers for the business.
- Florida (86) and New York (69) also show strong customer presence. These states could be considered high-value regional markets.
- Customer activity is not evenly distributed across states.

2.2 Question 2

Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT p.vehicle_maker,  
       COUNT(DISTINCT o.customer_id) AS  
       customer_count  
FROM order_t o  
JOIN product_t p USING (product_id)  
GROUP BY p.vehicle_maker  
ORDER BY customer_count DESC  
LIMIT 5;
```

Output:

Result: **Passed**

✓ Query 1

Query:

```
SELECT p.vehicle_maker,  
       COUNT(DISTINCT o.customer_id) AS customer_count  
FROM order_t o  
JOIN product_t p USING (product_id)  
GROUP BY p.vehicle_maker  
ORDER BY customer_count DESC  
LIMIT 5
```

Output:

Showing 5 rows

vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Fig 2.2 - Top Vehicle Makers by Customer Count

Observations and insights:

- Top 5 vehicle makers by customer count:
 - **Chevrolet — 83 customers**
 - **Ford — 63 customers**
 - **Toyota — 52 customers**
 - **Pontiac — 50 customers**
 - **Dodge — 50 customers**
- Chevrolet has the highest number of unique customers (83). This indicates strong popularity or demand for Chevrolet-related products.
- Ford and Toyota follow with 63 and 52 customers, respectively. The gap suggests Chevrolet has a noticeable lead, but the market remains competitive.
- Pontiac and Dodge both have 50 customers, showing similar levels of customer interest.

2.3 Question 3

Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT state, vehicle_maker, customer_count
FROM (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(DISTINCT o.customer_id) AS
customer_count,
        RANK() OVER (
            PARTITION BY c.state
            ORDER BY COUNT(DISTINCT
o.customer_id) DESC
        ) AS rnk
    FROM customer_t c
    JOIN order_t o USING (customer_id)
    JOIN product_t p USING (product_id)
    GROUP BY c.state, p.vehicle_maker
) ranked_data
WHERE rnk = 1;
```

Output:

```
Result: Passed

Query 1

Query:

SELECT state, vehicle_maker, customer_count
FROM (
  SELECT
    c.state,
    p.vehicle_maker,
    COUNT(DISTINCT o.customer_id) AS customer_count,
    RANK() OVER (
      PARTITION BY c.state
      ORDER BY COUNT(DISTINCT o.customer_id) DESC
    ) AS rnk
  FROM customer_t c
  JOIN order_t o USING (customer_id)
  JOIN product_t p USING (product_id)
  GROUP BY c.state, p.vehicle_maker
) ranked_data
WHERE rnk = 1
```

Fig 2.3.1 - Most Popular Vehicle Maker by State(Query)

Output:

Showing first 10 rows out of 143 rows

state	vehicle_maker	customer_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1
Arkansas	Pontiac	1
Arkansas	Mitsubishi	1
Arkansas	GMC	1
Arkansas	Chevrolet	1

Fig 2.3.2 - Most Popular Vehicle Maker by State

Observation and insights:

- The most popular vehicle maker in each state based on the number of unique customers who purchased products
 - **Alabama → Dodge (5 customers)**
 - **Alaska → Chevrolet (2 customers)**
 - **Arizona → Pontiac & Cadillac (tie with 3 customers)**
 - **Arkansas → Multiple brands tied with 1 customer**

- Different states show different preferred vehicle makers, indicating regional demand variations.
- States like Arkansas show several brands tied with very small counts (1 customer), suggesting:
 - Low demand volume
 - Highly fragmented market
- States like Alabama show a clear leader (Dodge with 5 customers), suggesting stronger brand preference.

2.4 Question 4

Find the overall average rating given by the customers. What is the average rating in each quarter?

Overall average rating for all quarters

Solution Query:

```
SELECT
    ROUND(AVG(rating_value), 2) AS overall_avg_rating
FROM (
    SELECT
        CASE
            WHEN customer_feedback = 'Very Bad' THEN 1
            WHEN customer_feedback = 'Bad' THEN 2
            WHEN customer_feedback = 'Okay' THEN 3
            WHEN customer_feedback = 'Good' THEN 4
            WHEN customer_feedback = 'Very Good' THEN 5
        END AS rating_value
    FROM order_t
) AS rating_table;
```

Output:

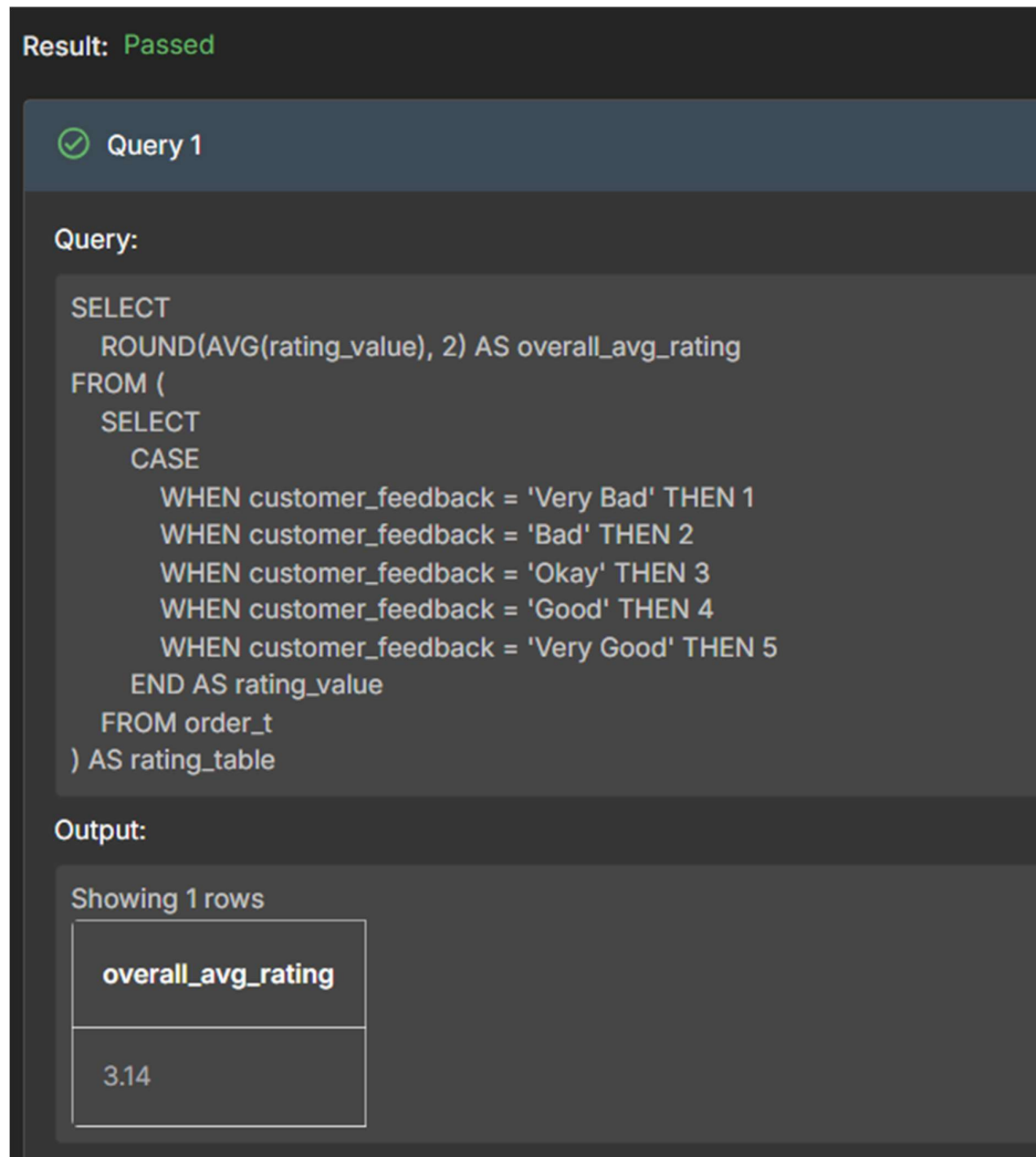


Fig 2.4.1 - Overall Average Customer Rating

Observation and insights:

- The average rating across all orders is 3.14.
- An average rating of 3.14 out of 5 indicates moderate customer satisfaction.

Average rating in each quarter

Solution Query:

```
SELECT
    quarter_number,
    ROUND(AVG(rating_value), 2) AS avg_rating_per_quarter
FROM (
    SELECT
        quarter_number,
        CASE
            WHEN customer_feedback = 'Very Bad' THEN 1
            WHEN customer_feedback = 'Bad' THEN 2
            WHEN customer_feedback = 'Okay' THEN 3
            WHEN customer_feedback = 'Good' THEN 4
            WHEN customer_feedback = 'Very Good' THEN 5
        END AS rating_value
    FROM order_t
) AS quarterly_rating_table
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: **Passed**

✓ Query 1

Query:

```
SELECT
  quarter_number,
  ROUND(AVG(rating_value), 2) AS avg_rating_per_quarter
FROM (
  SELECT
    quarter_number,
    CASE
      WHEN customer_feedback = 'Very Bad' THEN 1
      WHEN customer_feedback = 'Bad' THEN 2
      WHEN customer_feedback = 'Okay' THEN 3
      WHEN customer_feedback = 'Good' THEN 4
      WHEN customer_feedback = 'Very Good' THEN 5
    END AS rating_value
  FROM order_t
) AS quarterly_rating_table
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_rating_per_quarter
1	3.55
2	3.35
3	2.96
4	2.4

Fig 2.4.2 - Average rating in each quarter

Observation and insights:

- Quarter 1 has the highest satisfaction (3.55).
- Customer ratings gradually decline each quarter.
- Quarter 4 shows the lowest rating (2.40).
- Ratings drop consistently from Q1 → Q4. This suggests customer experience worsened throughout the year.
- Customers were relatively satisfied in Q1 and Q2 (above 3).
- By Q3 and Q4, ratings fall below 3, indicating dissatisfaction.
- The rating 2.40 is close to “Bad” feedback levels. This signals significant service, product, or delivery issues during the last quarter.

2.5 Question 5

Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
    quarter_number,

    ROUND(
        100.0 * SUM(CASE WHEN customer_feedback = 'Very
Bad' THEN 1 ELSE 0 END)
        / COUNT(*),
    4) AS very_bad_percentage,

    ROUND(
        100.0 * SUM(CASE WHEN customer_feedback = 'Bad'
THEN 1 ELSE 0 END)
        / COUNT(*),
    4) AS bad_percentage,

    ROUND(
        100.0 * SUM(CASE WHEN customer_feedback = 'Okay'
THEN 1 ELSE 0 END)
        / COUNT(*),
    4) AS okay_percentage,
```

```
ROUND(
    100.0 * SUM(CASE WHEN customer_feedback = 'Good'
THEN 1 ELSE 0 END)
    / COUNT(*),
4) AS good_percentage,

ROUND(
    100.0 * SUM(CASE WHEN customer_feedback = 'Very
Good' THEN 1 ELSE 0 END)
    / COUNT(*),
4) AS very_good_percentage

FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

```
Result: Passed

Query 1

Query:

SELECT
    quarter_number,

    ROUND(
        100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END)
        / COUNT(*),
        4) AS very_bad_percentage,

    ROUND(
        100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END)
        / COUNT(*),
        4) AS bad_percentage,

    ROUND(
        100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END)
        / COUNT(*),
        4) AS okay_percentage,

    ROUND(
        100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END)
        / COUNT(*),
        4) AS good_percentage,

    ROUND(
        100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END)
        / COUNT(*),
        4) AS very_good_percentage

FROM order_t

GROUP BY quarter_number

ORDER BY quarter_number
```

Fig 2.5.1 - Percentage distribution of feedback(Query)

Output:

Showing 4 rows

quarter_number	very_bad_percentage	bad_percentage	okay_percentage	good_percentage	very_good_percentage
1	10.9677	11.2903	19.0323	28.7097	30
2	14.8855	14.1221	20.229	22.1374	28.626
3	17.9039	22.7074	21.8341	20.9607	16.5939
4	30.6533	29.1457	20.1005	10.0503	10.0503

Fig 2.5.2 - Percentage distribution of feedback

Observation and insights:

- Among the four quarters, Q1 and Q2 shows very good rating with 30% and 28.626%.
- Q4 has the very bad rating at 30.6533%.
- Q1 has the highest positive feedback:
 - **Very Good (30%)**
 - **Good (28.7%)**
- Only about 22% negative feedback in Q1, indicating strong customer satisfaction.
- Positive ratings (Good + Very Good) decrease every quarter:
 - **Q1 → ~59%**
 - **Q2 → ~51%**
 - **Q3 → ~37%**
 - **Q4 → ~20%**
- Negative feedback (Very Bad + Bad) increases significantly:
 - **Q1 → ~22%**
 - **Q2 → ~29%**
 - **Q3 → ~40%**
 - **Q4 → ~60%**
- Q4 shows the worst customer sentiment:
 - **Very Bad: 30.65%**
 - **Bad: 29.15%**

2.6 Question 6

What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(DISTINCT order_id) AS order_count
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

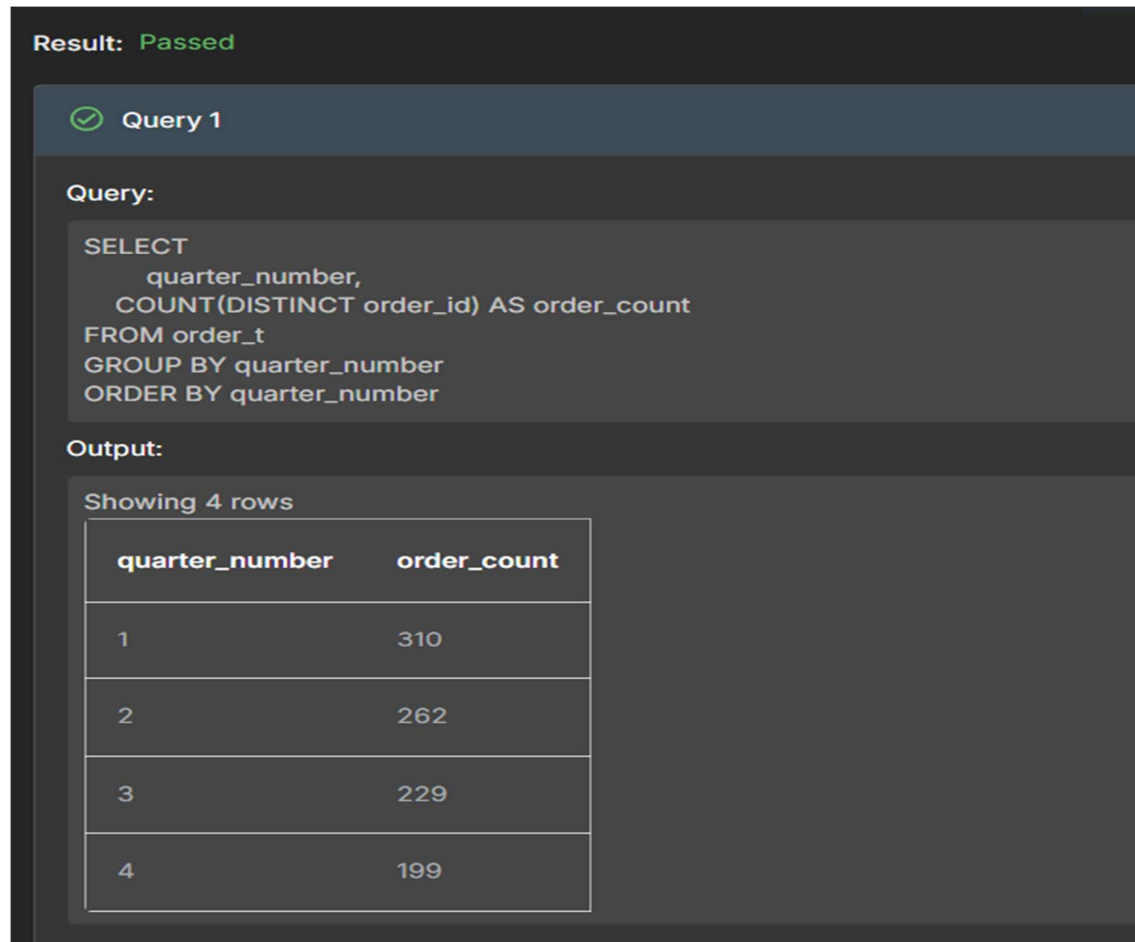


Fig 2.6 - Quarterly Orders

Observation and insights:

- Q1 has the highest number of orders (310).
- Orders consistently decline each quarter.
- Q4 has the lowest number of orders (199).
- Orders decrease steadily from Q1 → Q4.
- Total decline from Q1 to Q4 is 111 orders (~36% drop).
- Q1 shows the highest demand, indicating strong customer activity early in the year.
- The consistent drop suggests potential:
 - Customer dissatisfaction
 - Reduced repeat purchases
 - Competitive pressure

2.7 Question 7

Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
    quarter_number,
    net_revenue,
    LAG(net_revenue) OVER (ORDER BY quarter_number)
AS previous_quarter_revenue,

    ROUND(
        (net_revenue - LAG(net_revenue) OVER (ORDER
BY quarter_number))
        / LAG(net_revenue) OVER (ORDER BY
quarter_number) * 100, 2
    ) AS qoq_percentage_change

FROM (
    SELECT
        quarter_number,
        SUM(quantity * vehicle_price * (1 -
discount)) AS net_revenue
    FROM order_t
    GROUP BY quarter_number
) quarterly_revenue
ORDER BY quarter_number;
```

Output:

```
Result: Passed

Query 1

Query:

SELECT
  quarter_number,
  net_revenue,
  LAG(net_revenue) OVER (ORDER BY quarter_number) AS previous_quarter_revenue,

  ROUND(
    (net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))
    / LAG(net_revenue) OVER (ORDER BY quarter_number) * 100, 2
  ) AS qoq_percentage_change

FROM (
  SELECT
    quarter_number,
    SUM(quantity * vehicle_price * (1 - discount)) AS net_revenue
  FROM order_t
  GROUP BY quarter_number
) quarterly_revenue

ORDER BY quarter_number
```

Fig 2.7.1 - Quarterly Revenue(Query)

Output:

Showing 4 rows

quarter_number	net_revenue	previous_quarter_reve...	qoq_percentage_change
1	18032549.899600018		
2	13122995.7562	18032549.899600018	-27.23
3	8882298.8449	13122995.7562	-32.32
4	8573149.280599998	8882298.8449	-3.48

Fig 2.7.2 - Quarterly Revenue

Observation and insights:

- **Q1 generated the highest revenue (~\$18M).**
- Revenue **declined sharply in Q2 and Q3.**
- **Q4 stabilized slightly**, with only a small decline compared to Q3.

1. Significant Revenue Decline Across the Year

- Revenue dropped **over 50% from Q1 to Q4.**
- This indicates a **major slowdown in sales performance.**

2. Biggest Revenue Drop Occurred in Q3

- Q3 saw the **largest decline (-32.32%)** compared to Q2.
- This could indicate **demand reduction or operational challenges.**

3. Sales Stabilization in Q4

- The decline in Q4 was relatively small (**-3.48%**), suggesting the **business may have reached a revenue floor or stabilized demand.**

2.8 Question 8

What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(DISTINCT order_id) AS total_orders,
    ROUND(SUM(quantity * vehicle_price * (1 -
discount)), 2) AS total_net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

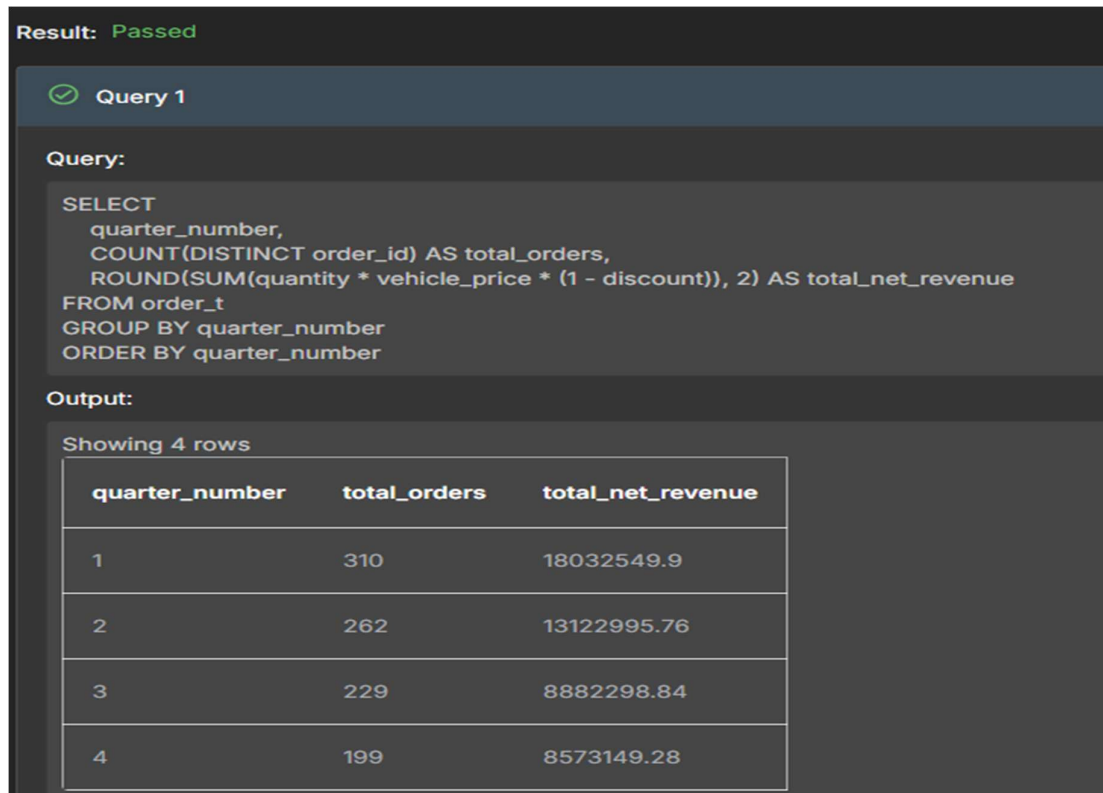


Fig 2.8 - Quarterly Orders and Total Net Revenue

Observation and insights:

- Q1 recorded the highest orders and revenue.
 - Both orders and revenue decline each quarter.
 - The drop is most noticeable between Q2 and Q3.
1. Sales Performance Declines Throughout the Year
 - Orders drop from 310 → 199 (about 36% decrease).
 - Revenue drops from \$18M → \$8.57M (about 52% decrease).
 2. Revenue Drops Faster Than Orders
 - Orders decrease gradually, but revenue declines more sharply.
 - This could indicate:
 - Higher discounts
 - Customers buying lower-priced vehicles
 - Reduced purchase quantities.

3. Strong Start but Weak End of the Year

- Q1 represents the peak performance period.
- Q4 is the weakest quarter in both demand and revenue.

2.9 Question 9

What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount), 2) AS avg_discount
FROM customer_t c
JOIN order_t o USING (customer_id)
GROUP BY c.credit_card_type;
```

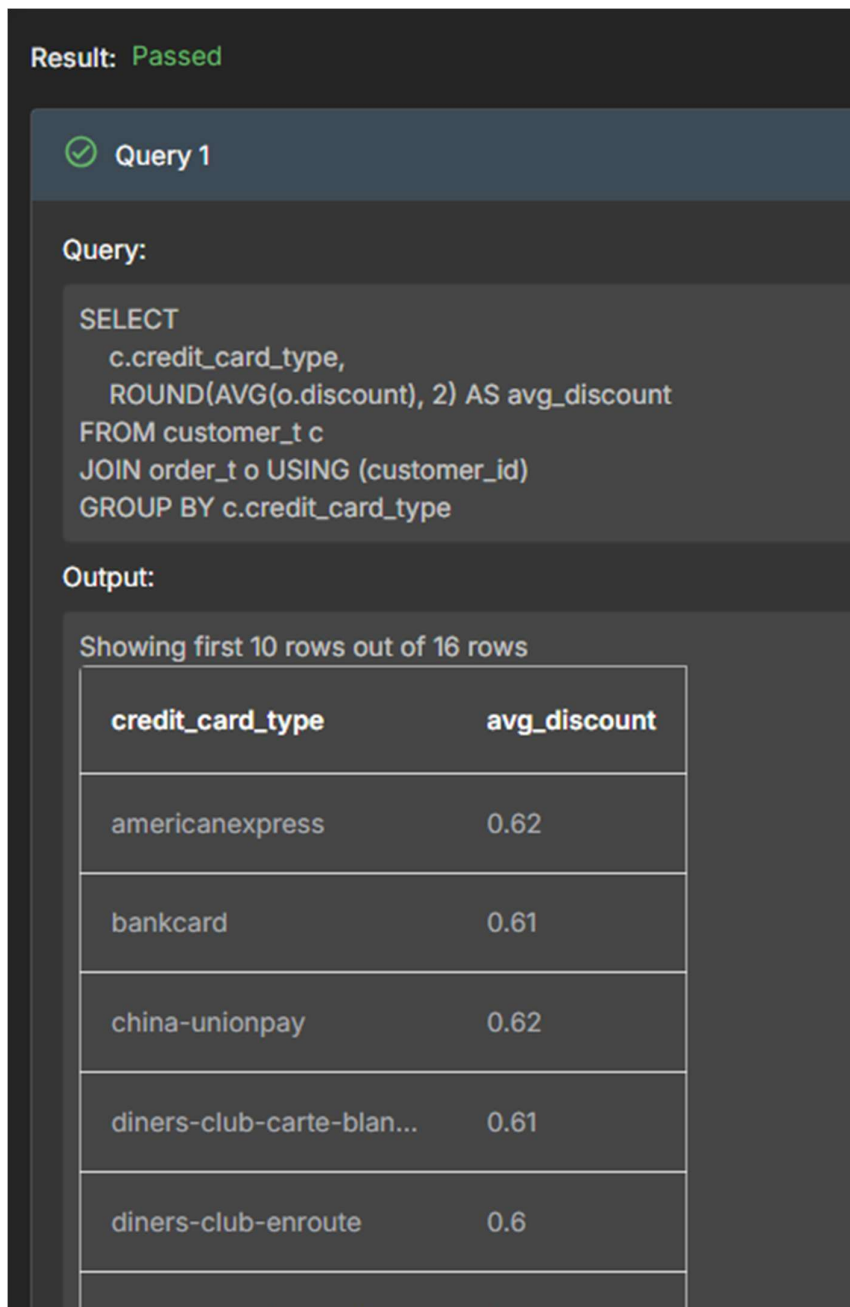


Fig 2.9 - Average Discount by Credit Card Type

Observation and insights:

- The average discount values are very similar across all credit card types.
- Discounts generally range between 0.58 and 0.64.

- The discount variation between card types is very small (~0.02 difference). This suggests discounts are not heavily influenced by the type of credit card used.
- The dataset includes many different credit card providers (16 types), indicating diverse payment preferences among customers.

2.10 Question 10

What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
    quarter_number,
    ROUND(AVG(julianday(ship_date) -
julianday(order_date)), 2) AS avg_shipping_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

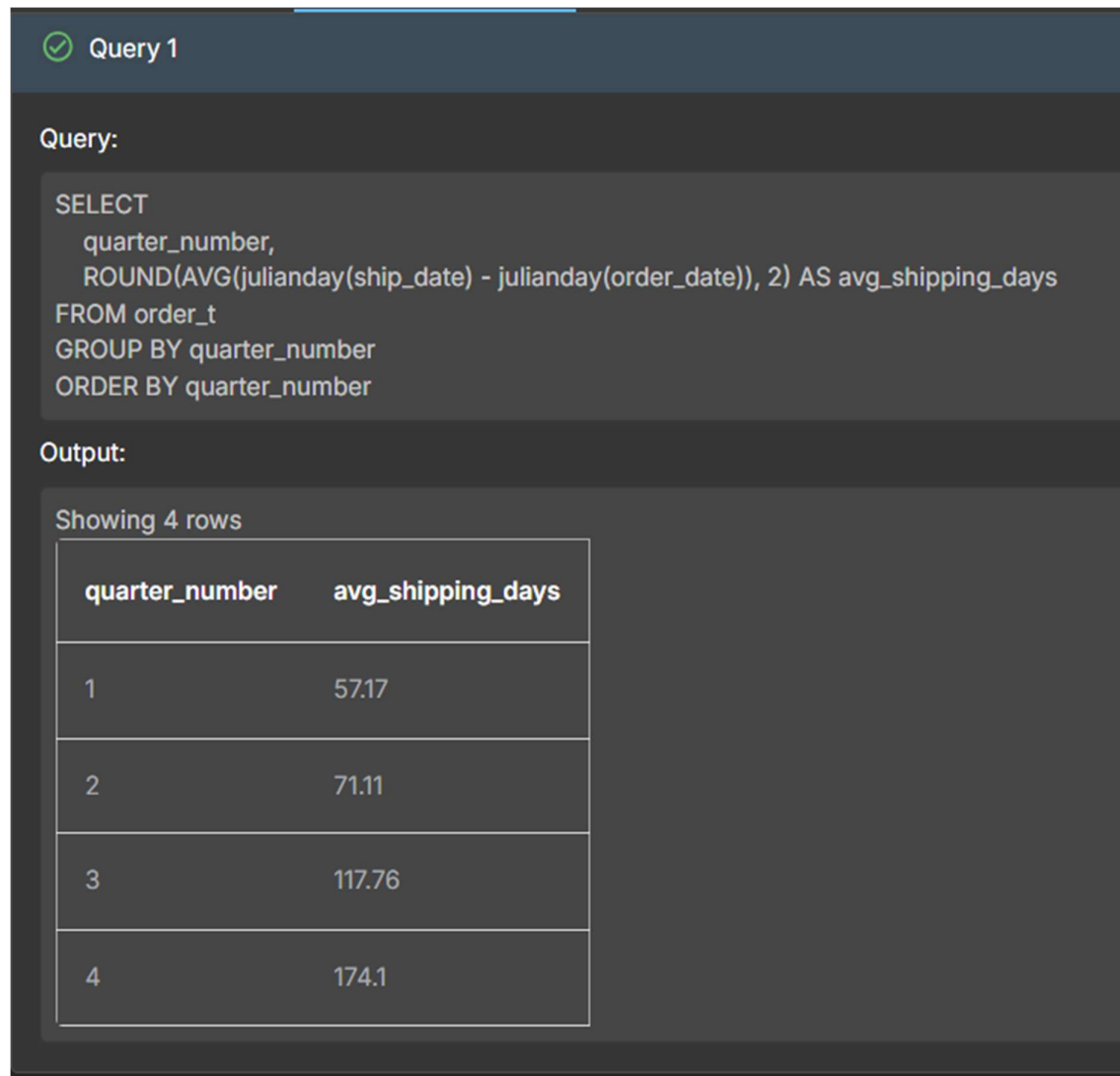


Fig 2.10 - Average Shipping Days by Quarter

Observation and insights:

- Shipping time increases significantly each quarter.
- Q4 shipping time is about 3× slower than Q1.
- Average shipping time rises from 57 days → 174 days. This indicates major logistical inefficiencies developing over time.
- Shipping time nearly doubles between Q2 and Q3.

- Q4 experiences the worst delay, suggesting supply chain or operational problems.
- Long shipping times can lead to:
 - Customer dissatisfaction
 - Higher negative feedback
 - Reduced repeat purchases
 - Declining sales and revenue

3. Business Metrics Overview:

Total Revenue	Total Orders	Total Customers	Average Rating
48610993.78	1000	994	3.14
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
8573149.28	199	97.96	44.1

Fig 3 - Key metrics

Observations:

1. Customer and order base

- With 1,000 orders coming from 994 customers, the repeat purchase rate is extremely low (0.6%), indicating low customer loyalty.

2. Revenue and Order Performance

- The company generated \$48.6M in total revenue from 1,000 orders.
- The last quarter contributed only ~\$8.57M, indicating a decline in revenue toward the end of the year

3. Customer Satisfaction

- The average rating is 3.14, which is slightly above neutral.
- Since less than half (44.1%) of customers give positive feedback, there is a significant dissatisfaction risk. The company should investigate the "Very Bad" and "Bad" segments to prevent churn.

4. Shipping Efficiency

- The average shipping time is 97.96 days, which is extremely long for order fulfilment, which is the primary driver for low rating.
- Earlier analysis showed shipping delays increase significantly over quarters.

5. Declining Business Trend

Earlier analyses showed:

- Orders decline each quarter
- Revenue declines each quarter
- Customer ratings decline each quarter
- Shipping delays increase each quarter

This suggests a **clear operational deterioration throughout the year.**

4. Business Recommendations:

1. Implement a Customer Retention Program

- Launch a "Loyalty Rewards" or "Referral Program." Since vehicle purchases are infrequent, focus on after-sales services (maintenance, parts, or insurance) to keep the 994 existing customers engaged with the brand.

2. Optimize Logistics & Supply Chain

- Conduct a root-cause analysis of the shipping pipeline.
- Implement Regional Distribution Centers in the top-performing states to reduce transit times. Reducing this to under 30 days should be the #1 priority to improve ratings.

3. Improve Customer Experience

- Focus on the "Dissatisfied" trend. Proactively reach out to customers who gave "Very Bad" or "Bad" feedback with service vouchers.
- Improve service, communication, and product quality. Improving the feedback loop will help stabilize the Quarter-over-Quarter % change.

4. Inventory Alignment by State

- Stock the "Most Preferred Maker" heavily in their respective states and reduce stock of slow-moving brands in those regions to save on storage costs and improve Sales Velocity.

5. Monitor Key KPIs

- Track regularly:
 - Shipping time
 - Customer rating
 - Repeat purchase rate
 - Revenue per quarter

Although the company generated strong total revenue (\$48.6M), operational inefficiencies—especially increasing shipping delays—likely drove declining customer satisfaction, reduced repeat purchases, and falling revenue toward the end of the year. Addressing logistics challenges and improving customer experience will be critical to reversing the decline and restoring sustainable growth.